Vicon ↔ 3DGS Camera Alignment

From COLMAP camera recovery to a saved coordinate transformation

FRAMES G: original 3DGS world V: Vicon world B: tracked marker body C: RGB optical camera

R_{AB} maps vectors from B into A. c: camera centre. b: marker-body position. s: metres / GS unit.

01 Recover the camera trajectory from COLMAP

Read the world-to-camera pose for each image from COLMAP images.bin.

$$p_C = R_{CG,k}\, p_G + t_{CG,k}$$

$$c_{G,k} = -R_{CG,k}^{T} t_{CG,k}, \qquad R_{GC,k} = R_{CG,k}^{T}$$

02 Match image times to Vicon poses

Use sensor-header timestamps; interpolate position linearly and orientation with SLERP.

$$\tau_k = t_k^{\mathrm{RGB}} + \Delta t$$

$$b_{V,k} = \mathrm{Interp}_p(\tau_k), \qquad R_{VB,k} = \mathrm{SLERP}_R(\tau_k)$$

03 Convert the marker-body pose to the camera pose

Fixed mount: d_B is the camera origin in B; R_{BC} maps camera axes into B.

$$c_{V,k} = b_{V,k} + R_{VB,k}\, d_B$$

$$R_{VC,k} = R_{VB,k}\, R_{BC}$$

04 Fit the scene scale, rotation and translation

Fit $\Theta = (s, R, t, R_{BC}, d_B, \Delta t)$; fix mounting/delay parameters if independently known.

$$c_{V,k} \simeq sR\, c_{G,k} + t, \qquad R_{VC,k} \simeq R\, R_{GC,k}$$

$$\min_{\Theta} \frac{1}{2} \sum_{k \in \mathcal{F}} \sum_{i=1}^{3} \left[\rho\left(\frac{e_{p,k,i}^2}{\sigma_p^2} \right) + \rho\left(\frac{e_{R,k,i}^2}{\sigma_R^2} \right) \right]$$

05 Validate the alignment on excluded poses

Fit on selected time blocks; evaluate excluded blocks and check independent physical dimensions.

$$\mathbf{e}_{p,k} = sR\, c_{G,k} + t - c_{V,k}, \qquad E_{p,k} = \|\mathbf{e}_{p,k}\|_2$$

$$\mathbf{e}_{R,k} = \mathrm{Log}\left(R_{VC,k}^{T} R\, R_{GC,k}\right)^{\vee}, \qquad E_{\theta,k} = \frac{180}{\pi} \|\mathbf{e}_{R,k}\|_2$$

06 Save the calibration; use its inverse for rendering

Store $s, R, t, R_{BC}, d_B, \Delta t$. Here $R = R_{VG}$. The original PLY can stay unchanged.

$$c_G = \frac{1}{s} R^T (c_V - t) = \frac{1}{s} R^T (b_V + R_{VB} d_B - t)$$

$$R_{GC} = R^T R_{VC} = R^T R_{VB} R_{BC}$$

FIT DETAILS $\mathcal{F}$: fitting poses; ρ: soft-L1 loss; $\sigma_p = 0.02$ m; $\sigma_R = 1^\circ$ (converted to radians).

$\mathrm{Log}(\cdot)^{\vee}$ converts a relative rotation matrix to a rotation vector in radians.

Current calibration: mount and alignment were estimated jointly. Independent mounting verification is still needed.